

MATH 2123 - MATHEMATICS III

2022 Exam Paper: Beginner Guide + Worked Solutions

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Designed for a student who is NOT yet an expert.

Each available set includes: Topic -> Basics -> Problem-solving procedure -> Full solution -> Exam tips.

Important source note

You uploaded page 1 of 3 and page 2 of 3 of the 2022 paper.

Questions 1-7 are visible and are fully explained in this guide.

Question 8 is on the missing third page, so Set 8 cannot be solved accurately yet. A placeholder is included instead of guessing.

How to use this guide

1. Read the Basics first; they explain what the formulas mean.
2. Memorize the Procedure as a reusable recipe, not just the numerical result.
3. Follow the worked solution line by line and check the reason for each step.
4. Before the exam, revise the green Final Answer boxes and the rapid-revision pages.

Paper map - what all 8 sets are really testing

Set	Main topic
1	Space curves (T, N, curvature, normal plane); solenoidal vs rotational vector fields
2	Directional derivatives; solenoidal/conservative fields; work along a parameterized curve
3	Surface flux; piecewise line integral; divergence volume integral
4	Matrix echelon/canonical/normal forms and rank; parameter-dependent linear system
5	Laplace transform by differentiation; periodic functions; convolution theorem
6	Inverse Laplace; ODE by Laplace; integral equation by Laplace
7	Beam deflection; orthogonal matrices; eigenvalues/eigenvectors
8	Not visible - page 3 of the paper was not uploaded

The 2022 paper places vector calculus and matrices in Section A, then Laplace transforms, beam deflection, and matrix/eigen topics in Section B.

SET 1 - Space Curves and Solenoidal/Rotational Fields

What this set tests

- 1(a) Unit tangent T , unit normal N , curvature/radius of curvature, and the normal plane of a space curve.
- 1(b) How divergence and curl tell you whether a vector field is solenoidal and rotational.

Basics you need

- For $r(t)$, velocity is $r'(t)$, speed is $|r'(t)|$, and unit tangent is $T=r'/|r'|$.
- The principal unit normal is $N=T'/|T'|$, provided T' is nonzero.
- Curvature $\kappa=|T'|/|r'|$ and radius of curvature $\rho=1/\kappa$.
- The normal plane is perpendicular to T . Through point r_0 , its equation is $T_0 \cdot (r-r_0)=0$.
- A vector field is solenoidal if $\text{div } F=0$. It is rotational if $\text{curl } F$ is not zero.

Reusable procedure

Space curve: differentiate $r \rightarrow$ find speed $\rightarrow T \rightarrow$ differentiate $T \rightarrow N \rightarrow$ curvature $\rightarrow$ evaluate at required $t \rightarrow$ write plane.

Field classification: compute divergence and curl separately. $\text{div}=0$ means no net source/sink; curl nonzero means local spinning.

1(a) $r(t)=2 \cos t \mathbf{i} + 2 \sin t \mathbf{j} + \mathbf{k}$

Write the curve in coordinate form: $r(t)=(2 \cos t, 2 \sin t, 1)$. The z -coordinate is constant, so the path is a horizontal circle.

$$r'(t)=(-2 \sin t, 2 \cos t, 0)$$

$$|r'(t)|=\sqrt{4 \sin^2 t + 4 \cos^2 t}=2$$

$$T(t)=(-\sin t, \cos t, 0)$$

$$T'(t)=(-\cos t, -\sin t, 0), \quad |T'|=1$$

$$N(t)=(-\cos t, -\sin t, 0)$$

$$\kappa=|T'|/|r'|=1/2, \quad \rho=1/\kappa=2$$

At $t=\pi/2$:

$$r_0=(0, 2, 1), \quad T_0=(-1, 0, 0), \quad N_0=(0, -1, 0)$$

The normal plane is perpendicular to T_0 :

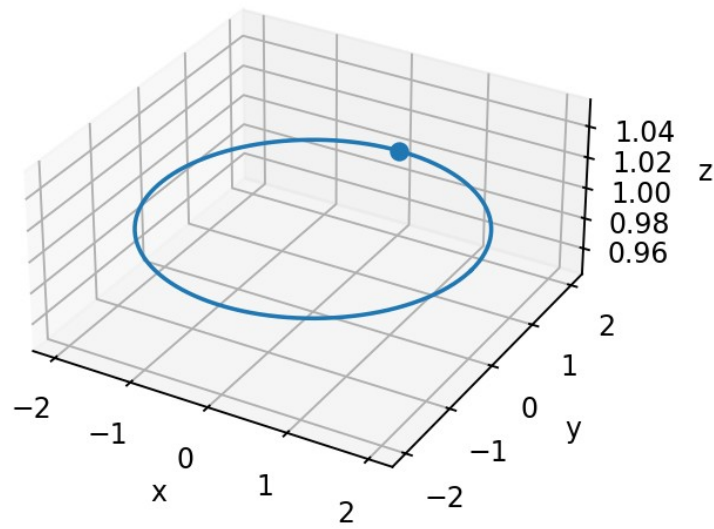
$$T_0 \cdot [(x, y, z) - (0, 2, 1)] = 0$$

$$-(x-0)=0 \rightarrow x=0$$

Final answer

$T=(-\sin t, \cos t, 0)$, $N=(-\cos t, -\sin t, 0)$, $\kappa=1/2$, $\rho=2$. At $t=\pi/2$ the normal plane is $x=0$.

$$r(t) = (2 \cos t, 2 \sin t, 1): \text{circle } x^2 + y^2 = 4 \text{ at } z=1$$



The curve is a circle of radius 2 in the plane $z=1$.

1(b) How to identify a field that is both solenoidal and rotational

Suppose $F=(P,Q,R)$. Two different tests are used:

$$\text{div } F = P_x + Q_y + R_z$$

$$\text{curl } F = \nabla \times F$$

A field is both solenoidal and rotational when:

$$\text{div } F = 0 \quad \text{AND} \quad \text{curl } F \neq 0$$

Example: $F=(-y, x, 0)$.

$$\text{div } F = d(-y)/dx + d(x)/dy + 0 = 0$$

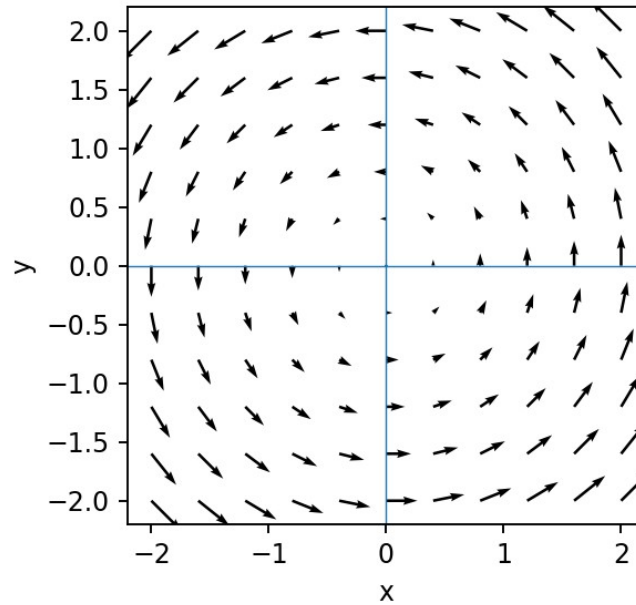
$$\text{curl } F = (0, 0, d(x)/dx - d(-y)/dy) = (0, 0, 2) \neq 0$$

So this example has no source/sink but it does have circulation (local rotation).

Final answer

Identification rule: solenoidal $\rightarrow \text{div } F=0$; rotational $\rightarrow \text{curl } F$ is nonzero. Both conditions must hold simultaneously.

Example $F=(-y, x)$: circular flow, $\text{div } F=0$, $\text{curl } F=2\mathbf{k}$



A simple rotational-but-solenoidal field: arrows circulate around the origin.

Common mistake / exam note

Do not confuse "rotational" with "irrotational". Irrotational means $\text{curl } F=0$; rotational means $\text{curl } F$ is not zero.

SET 2 - Directional Derivatives, Solenoidal Fields and Work

What this set tests

- 2(a) Use the gradient to determine unknown constants from the maximum directional derivative.
- 2(b) Definitions of solenoidal/conservative fields and a product-divergence calculation.
- 2(c) Work done by a vector force along a parametrically defined curve.

Basics you need

- For scalar f , $\text{grad } f = (f_x, f_y, f_z)$. The maximum directional derivative is $|\text{grad } f|$ and points along $\text{grad } f$.
- If the maximum direction is parallel to one coordinate axis, the other gradient components must be zero.
- Solenoidal means $\text{div } F = 0$. Conservative means $F = \text{grad } \phi$ for some scalar ϕ ; in a simply connected region, $\text{curl } F = 0$ is the standard test.
- Work along a curve C : $W = \int_C F \cdot dr$. With $r(t)$, $W = \int F(r(t)) \cdot r'(t) dt$.

Reusable procedure

Directional derivative: compute grad $\rightarrow$ substitute point $\rightarrow$ force unwanted components to zero $\rightarrow$ use magnitude $\rightarrow$ evaluate desired directional derivative.

Solenoidal product: use $\text{div}(\phi A) = \text{grad}(\phi) \cdot A + \phi \text{div } A$; cancellation is often the point of the question.

Work: parameterize $\rightarrow$ differentiate $\rightarrow$ substitute $x(t), y(t), z(t)$ into $F \rightarrow$ dot product $\rightarrow$ integrate over t .

2(a) Find a, b, c from the maximum directional derivative

Given $f = ayz^2 + bzx + cx^2y^3$. At $(2, -1, 1)$, the maximum directional derivative has magnitude 64 in a direction parallel to the y -axis.

$$f_x = bz + 2cxy^3$$

$$f_y = az^2 + 3cx^2y^2$$

$$f_z = 2ayz + bx$$

At $(2, -1, 1)$:

$$\text{grad } f = (b - 4c, a + 12c, -2a + 2b)$$

For $\text{grad } f$ to be parallel to the y -axis:

$$b - 4c = 0 \rightarrow b = 4c$$

$$-2a + 2b = 0 \rightarrow a = b = 4c$$

The y -component becomes $a + 12c = 16c$. Under the usual exam convention that the maximum direction is $+y$:

$$16c = 64 \rightarrow c = 4 \rightarrow a = b = 16$$

For the same f , the directional derivative along $+x$ at $(1, -1, 2)$ is f_x there:

$$D_i f = f_x = bz + 2cxy^3 = 16(2) + 2(4)(1)(-1) = 24$$

Final answer

Expected $+y$ -axis answer: $a=16, b=16, c=4$; directional derivative along $+x$ at $(1, -1, 2) = 24$.

Common mistake / exam note

The print appears to contain a small typo in the last sentence (it looks like " $f=c$ "). The meaningful interpretation is the directional derivative of the same f . Also, if "parallel" is interpreted as either $+y$ or $-y$, the sign-reversed set $a=b=-16, c=-4$ gives maximum magnitude 64 and the x -direction derivative -24.

2(b) Show A is not solenoidal but B=xyz² A is solenoidal

The paper gives:

$$A = (2x^2 + 8xy^2z)i + (3x^3y - 3xy)j - (4y^2z^2 + 2x^3z)k$$

First compute the divergence:

$$\begin{aligned}\operatorname{div} A &= (4x + 8y^2z) + (3x^3 - 3x) - (8y^2z + 2x^3) \\ &= x + x^3 = x(1 + x^2)\end{aligned}$$

This is not identically zero, so A is not solenoidal.

Now let $\phi = xyz^2$ and $B = \phi A$. Use the product rule:

$$\begin{aligned}\operatorname{div}(\phi A) &= \operatorname{grad}(\phi) \cdot A + \phi \operatorname{div} A \\ \operatorname{grad}(\phi) &= (yz^2, xz^2, 2xyz)\end{aligned}$$

Compute the dot product and simplify:

$$\begin{aligned}\operatorname{grad}(\phi) \cdot A &= -x^2 y z^2 (1 + x^2) \\ \phi \operatorname{div} A &= xyz^2 \cdot x(1 + x^2) = x^2 y z^2 (1 + x^2) \\ \text{Therefore } \operatorname{div} B &= 0\end{aligned}$$

Final answer

A is not solenoidal because $\operatorname{div} A = x(1 + x^2)$. $B = xyz^2 A$ is solenoidal because $\operatorname{div} B = 0$.

Definitions requested in the question

Solenoidal vector field: a vector field with divergence zero throughout the region.

Conservative vector field: a field F for which there exists a scalar potential ϕ such that $F = \operatorname{grad} \phi$. In a simply connected region, $\operatorname{curl} F = 0$ is the usual test.

2(c) Work done along $x = t^2 + 1$, $y = 2t^2$, $z = t^3$, $1 \leq t \leq 2$

$F = 3xy i - 5z j + 10x k$.

$$r(t) = (t^2 + 1, 2t^2, t^3)$$

$$r'(t) = (2t, 4t, 3t^2)$$

$$F(r(t)) = (6t^2(t^2 + 1), -5t^3, 10(t^2 + 1))$$

$$F \cdot r' = 12t^5 + 10t^4 + 12t^3 + 30t^2$$

$$W = \int_1^2 (12t^5 + 10t^4 + 12t^3 + 30t^2) dt$$

$$= [2t^6 + 2t^5 + 3t^4 + 10t^3]_1^2$$

$$= 320 - 17 = 303$$

Final answer

Work done = 303 units.

SET 3 - Surface Flux, Piecewise Line Integral and Divergence Integral

What this set tests

- 3(a) Flux through a plane surface in the first octant.
- 3(b) Work/line integral along two straight-line path segments.
- 3(c) Integrate the divergence of a field over a rectangular box.

Basics you need

- For $z=f(x,y)$, an upward vector area element is $(-f_x, -f_y, 1) dx dy$.
- A line integral on a piecewise path is the sum over each segment.
- For $F=(P,Q,R)$, $\text{div } F = P_x + Q_y + R_z$. Compute this before the triple integral.

Reusable procedure

Plane flux: solve for $z \rightarrow$ find vector area element $\rightarrow$ project onto xy -plane $\rightarrow$ set triangular limits $\rightarrow$ integrate $F \cdot dS$.

Piecewise line: parameterize segment 1 and segment 2 separately $\rightarrow$ calculate each integral $\rightarrow$ add.

Divergence volume integral: compute $\text{div } F \rightarrow$ write box limits $\rightarrow$ integrate, using constant side lengths when possible.

3(a) Flux through $2x+2y+z=4$ in the first octant

Given $F=(3z)i - 2xy j - 4x k$. Solve the plane for z :

$$z=4-2x-2y$$

For $r(x,y)=(x,y,4-2x-2y)$,

$$r_x \times r_y = (2, 2, 1) \quad [\text{normal with positive } z\text{-component}]$$

The projected triangular region is $x \geq 0$, $y \geq 0$, $x+y \leq 2$, so $0 \leq x \leq 2$ and $0 \leq y \leq 2-x$.

$$F \cdot (2, 2, 1) = 6z - 4xy - 4x$$

$$= 6(4-2x-2y) - 4xy - 4x$$

$$= 24 - 16x - 12y - 4xy$$

$$\text{Flux} = \int_0^2 \int_0^{2-x} (24 - 16x - 12y - 4xy) dy dx$$

$$= 8$$

Final answer

Flux = 8 for the normal pointing away from the origin / with positive z -component. The opposite orientation gives -8.

Common mistake / exam note

A surface-flux sign depends on orientation. If the question does not explicitly specify it, state which normal you choose.

3(b) Line integral along $(0,0) \rightarrow (2,0) \rightarrow (3,2)$

$F=(2x+y)i+(3y-x)j$.

Segment C1: $(0,0)$ to $(2,0)$

$$x=t, y=0, 0 \leq t \leq 2; \quad dx=dt, dy=0$$

$$\text{Integral_C1} = \int_0^2 (2t) dt = 4$$

Segment C2: (2,0) to (3,2)

$$x=2+t, y=2t, 0 \leq t \leq 1; \quad dx=dt, \quad dy=2dt$$

$$P=2x+y=4+4t, \quad Q=3y-x=5t-2$$

$$P dx + Q dy = [(4+4t)+2(5t-2)]dt = 14t dt$$

$$\text{Integral_C2} = \int_0^1 14t dt = 7$$

Final answer

Total line integral = $4+7 = 11$.

3(c) Triple integral of div F over $1 \leq x, y, z \leq 3$

$$F = (2x^3 - 3z)\mathbf{i} - 2xy\mathbf{j} - 4x\mathbf{k}.$$

$$\begin{aligned} \text{div } F &= \frac{d}{dx}(2x^3 - 3z) + \frac{d}{dy}(-2xy) + \frac{d}{dz}(-4x) \\ &= 6x^2 - 2x \end{aligned}$$

The y and z intervals both have length 2, so:

$$\begin{aligned} \text{Integral} &= 4 * \int_1^3 (6x^2 - 2x) dx \\ &= 4[2x^3 - x^2]_1^3 \\ &= 4[(54 - 9) - (2 - 1)] = 4(44) = 176 \end{aligned}$$

Final answer

The value of the volume integral is 176.

SET 4 - Rank, Canonical/Normal Form and a Parameter System

What this set tests

4(a) Row-echelon form, row canonical form (RREF), normal form, and matrix rank.

4(b) Decide when a system has a unique solution, infinitely many solutions, or no solution as k changes.

Basics you need

- Row echelon form: pivots move to the right as you go down; entries below each pivot are zero.
- Row canonical form/RREF: every pivot is 1 and is the only nonzero entry in its pivot column.
- Rank = number of pivots/nonzero rows in echelon form.
- Normal form under row and column operations is $[I_r \ 0; \ 0 \ 0]$. For a 3×4 rank-3 matrix, normal form is $[I_3 \ 0]$.
- For a square system $Ax=b$: $\det A \neq 0 \rightarrow$ unique solution. If $\det A=0$, compare $\text{rank}(A)$ with $\text{rank}([A|b])$.

Reusable procedure

Matrix forms: use row operations to get echelon $\rightarrow$ continue to RREF $\rightarrow$ count pivots $\rightarrow$ use column operations to clear non-pivot columns for normal form.

Parameter system: compute determinant $\rightarrow$ solve critical k values $\rightarrow$ at each critical value compare ranks $\rightarrow$ classify.

4(a) Reduce A and find rank

$$A = \begin{bmatrix} 0 & 1 & 0 & -2 \\ -4 & 2 & 1 & 1 \\ 2 & 0 & 1 & 0 \end{bmatrix}$$

A convenient sequence is: swap $R1 \leftrightarrow R3$, then eliminate below the first pivot, then place the y -pivot above the last row.

$$R1 \leftrightarrow R3$$

$$R2 \leftarrow R2 + 2R1$$

$$R2 \leftrightarrow R3$$

$$R3 \leftarrow R3 - 2R2$$

$$\text{Echelon form} = \begin{bmatrix} 2 & 0 & 1 & 0 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 3 & 5 \end{bmatrix}$$

There are three pivots, so $\text{rank } A = 3$.

Continue to RREF:

$$R1 \leftarrow R1/2, \quad R3 \leftarrow R3/3, \quad R1 \leftarrow R1 - (1/2)R3$$

$$\text{Row canonical form} = \begin{bmatrix} 1 & 0 & 0 & -5/6 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 5/3 \end{bmatrix}$$

Now use a column operation to clear the final column:

$$C4 \leftarrow C4 + (5/6)C1 + 2C2 - (5/3)C3$$

$$\text{Normal form} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} = [I_3 \mid 0]$$

Final answer

Rank $A=3$. One echelon form is $\begin{bmatrix} 2 & 0 & 1 & 0 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 3 & 5 \end{bmatrix}$. RREF is $\begin{bmatrix} 1 & 0 & 0 & -5/6 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 5/3 \end{bmatrix}$.

Normal form is $[I_3 \mid 0]$.

4(b) Classify the system depending on k

Coefficient matrix and RHS:

$$A(k) = \begin{bmatrix} 2 & 3 & k \\ 1 & 6 & 2k+3 \\ 2 & 3 & 2 \end{bmatrix}, \quad b = (1, 2, 1)^T$$

$$\det A(k) = -9(k-2)$$

Therefore, for $k \neq 2$ the determinant is nonzero and the system has a unique solution.

Solving gives the remarkably simple result:

$$x=0, \quad y=1/3, \quad z=0 \quad (\text{for every } k \neq 2)$$

At $k=2$, rows 1 and 3 of the augmented system are identical, and $\text{rank } A = \text{rank}[A|b] = 2 < 3$. Therefore there are infinitely many solutions.

$$\text{At } k=2, \text{ let } z=t. \text{ Then } x=t, \quad y=1/3 - (4/3)t.$$

Final answer

Unique solution for $k \neq 2$: $(x,y,z)=(0,1/3,0)$. Infinite solutions for $k=2$: $(x,y,z)=(t, 1/3-4t/3, t)$. No value of k gives no solution.

SET 5 - Laplace Differentiation, Periodic Functions and Convolution

What this set tests

- 5(a) Find $L\{t^3 \sin t\}$ using differentiation with respect to s , then evaluate an improper integral.
- 5(b) Laplace transform of a periodic piecewise function.
- 5(c) Convolution theorem for an inverse Laplace transform.

Basics you need

- Definition: $L\{f(t)\} = \int_0^\infty e^{-st} f(t) dt$, when the integral converges.
- If $F(s) = L\{f(t)\}$, then $L\{t^n f(t)\} = (-1)^n \frac{d^n F}{ds^n}$.
- If f has period T : $L\{f\} = \frac{\int_0^T e^{-st} f(t) dt}{1 - e^{-sT}}$.
- Convolution: $L^{-1}\{F(s)G(s)\} = \int_0^t f(\tau)g(t-\tau) d\tau$.

Reusable procedure

Multiplication by t^n : start from a known simple transform $\rightarrow$ differentiate n times $\rightarrow$ apply $(-1)^n$.

Periodic function: integrate only over one period, piece by piece $\rightarrow$ divide by $1 - e^{-sT}$.

Convolution: split the s -expression into two known transforms $\rightarrow$ write convolution integral $\rightarrow$ evaluate.

5(a) $L\{t^3 \sin t\}$ and the improper integral

Start with:

$$L\{\sin t\} = 1/(s^2 + 1)$$

Use $L\{t^3 f(t)\} = (-1)^3 \frac{d^3}{ds^3} F(s)$:

$$L\{t^3 \sin t\} = 24s(s^2 - 1)/(s^2 + 1)^4$$

Now observe:

$$\begin{aligned} \int_0^\infty t^3 e^{-t} \sin t \, dt &= L\{t^3 \sin t\} \text{ evaluated at } s=1 \\ &= 24(1)(1-1)/(1+1)^4 = 0 \end{aligned}$$

Final answer

$L\{t^3 \sin t\} = 24s(s^2 - 1)/(s^2 + 1)^4$, and $\int_0^\infty t^3 e^{-t} \sin t \, dt = 0$.

5(b) Periodic $F(t)$ with period 4

$F(t) = 3t$ for $0 < t < 2$, and $F(t) = 6$ for $2 < t < 4$; $F(t+4) = F(t)$.

$$L\{F\} = \left[\int_0^2 3t e^{-st} dt + \int_2^4 6e^{-st} dt \right] / (1 - e^{-4s})$$

The two pieces simplify to:

$$\begin{aligned} &\int_0^2 3t e^{-st} dt + \int_2^4 6e^{-st} dt \\ &= 3(1 - e^{-2s})/s^2 - 6e^{-4s}/s \end{aligned}$$

Final answer

$L\{F(t)\} = [3(1 - e^{-2s})/s^2 - 6e^{-4s}/s] / (1 - e^{-4s})$.

Common mistake / exam note

For a periodic function, do NOT integrate from 0 to infinity directly piece by piece. Use one period and divide by $1 - e^{-sT}$.

5(c) Use convolution to find $L^{-1}\{1/[s^2(s+1)^2]\}$

$$1/[s^2(s+1)^2] = (1/s^2) * (1/(s+1)^2)$$

$$L^{-1}\{1/s^2\} = t, \quad L^{-1}\{1/(s+1)^2\} = t e^{-t}$$

$$h(t) = \int_0^t \tau (t-\tau) e^{-(t-\tau)} d\tau$$

Put $u = t - \tau$ and expand, or integrate directly. The simplification is:

$$h(t) = t - 2 + (t+2)e^{-t}$$

Final answer

$$L^{-1}\{1/[s^2(s+1)^2]\} = t - 2 + (t+2)e^{-t}.$$

SET 6 - Inverse Laplace, ODE and Integral Equation

What this set tests

- 6(a) Partial fractions and standard inverse Laplace transforms.
- 6(b) Solve a second-order initial-value ODE with Laplace transforms.
- 6(c) Recognize a convolution inside an integral equation and solve algebraically in the s-domain.

Basics you need

- Use partial fractions after factoring the denominator.
- $L\{y'\} = sY - y(0)$, and $L\{y''\} = s^2Y - sy(0) - y'(0)$.
- A convolution integral $\int_0^t g(x)F(t-x)dx$ transforms into $G(s)Fbar(s)$.

Reusable procedure

Inverse rational transform: factor $\rightarrow$ partial fractions $\rightarrow$ match each term to a transform table.

ODE: transform $\rightarrow$ substitute initial conditions immediately $\rightarrow$ solve for $Y(s)$ $\rightarrow$ simplify $\rightarrow$ inverse transform.

Integral equation: identify convolution $\rightarrow$ transform both sides $\rightarrow$ isolate the unknown transform $\rightarrow$ invert.

6(a) Inverse Laplace of $(s+4)/[s(s-1)(s^2+4)]$

Partial fractions reduce very cleanly:

$$(s+4)/[s(s-1)(s^2+4)] = -1/s + 1/(s-1) - 1/(s^2+4)$$

$$L^{-1}\{-1/s\} = -1$$

$$L^{-1}\{1/(s-1)\} = e^t$$

$$L^{-1}\{1/(s^2+4)\} = (1/2)\sin 2t$$

Final answer

$$f(t) = e^t - 1 - (1/2)\sin 2t.$$

6(b) Solve $y''+4y'+4y=4e^{-2t}$, $y(0)=-1$, $y'(0)=4$

Let $L\{y\} = Y(s)$.

$$(s^2Y - sy(0) - y'(0)) + 4(sY - y(0)) + 4Y = 4/(s+2)$$

Insert $y(0)=-1$ and $y'(0)=4$:

$$(s+2)^2 Y + s = 4/(s+2)$$

$$Y = 4/(s+2)^3 - s/(s+2)^2$$

$$= 4/(s+2)^3 - 1/(s+2) + 2/(s+2)^2$$

Invert term by term:

$$L^{-1}\{4/(s+2)^3\} = 2t^2 e^{-2t}$$

$$L^{-1}\{-1/(s+2)\} = -e^{-2t}$$

$$L^{-1}\{2/(s+2)^2\} = 2t e^{-2t}$$

Final answer

$$y(t) = e^{-2t}(2t^2 + 2t - 1).$$

6(c) Integral equation $F(t)=1+2 \int_0^t e^{-2x}F(t-x)dx$

The integral is a convolution of e^{-2t} and $F(t)$. Let the Laplace transform of F be $Y(s)$.

$$Y(s)=1/s + 2[1/(s+2)]Y(s)$$

$$Y(s)[1-2/(s+2)] = 1/s$$

$$Y(s)*s/(s+2)=1/s$$

$$Y(s)=(s+2)/s^2 = 1/s + 2/s^2$$

Final answer

$$F(t)=1+2t.$$

SET 7 - Beam Deflection, Orthogonal Matrices and Eigenvalues

What this set tests

- 7(a) Deflection of a simply supported beam under a uniform load.
- 7(b) Prove a matrix of direction cosines of mutually perpendicular lines is orthogonal.
- 7(c) Find eigenvalues and the eigenvector for the largest eigenvalue.

Basics you need

- For a simply supported beam, vertical deflection is zero at both supports. A uniform load gives equal reactions at the ends.
- Beam relation: $EI y'' = M(x)$ under the common sign convention. E is Young's modulus and I is second moment of area.
- A square matrix A is orthogonal if $AA^T = I$ (equivalently $A^{-1} = A^T$).
- Eigenvalues solve $\det(A - \lambda I) = 0$; eigenvectors solve $(A - \lambda I)v = 0$.

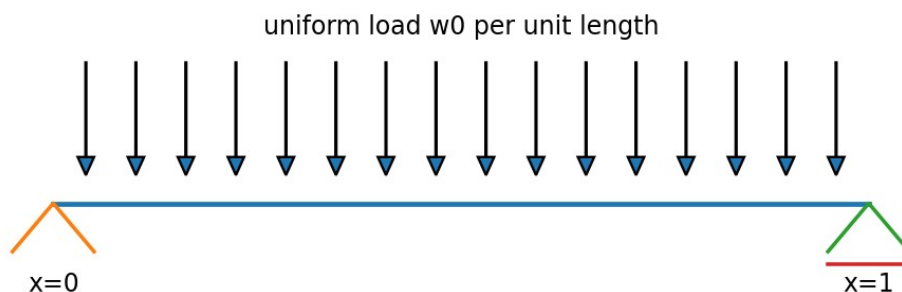
Reusable procedure

Beam: find reactions $\rightarrow$ write bending moment at section $x \rightarrow$ integrate twice $\rightarrow$ apply $y(0) = y(L) = 0 \rightarrow$ interpret sign.

Orthogonal proof: show each row is a unit vector and different rows have zero dot product $\rightarrow AA^T = I$.

Eigenproblem: characteristic determinant $\rightarrow$ roots $\rightarrow$ solve one linear equation for each eigenvector.

7(a) Beam hinged at $x=0$ and $x=1$ with uniform load w_0



Simply supported beam of unit length carrying a uniform load w_0 .

Total load $= w_0 \cdot 1 = w_0$, so by symmetry the reactions are $R_A = R_B = w_0/2$.

At a section x from the left support, the bending moment is:

$$M(x) = R_A x - (w_0 x)(x/2) = (w_0/2)(x - x^2)$$

Using $EI y'' = M(x)$ with y positive upward:

$$EI y' = w_0 x^2/4 - w_0 x^3/6 + C_1$$

$$EI y = w_0 x^3/12 - w_0 x^4/24 + C_1 x + C_2$$

Boundary conditions $y(0) = 0$ and $y(1) = 0$ give $C_2 = 0$ and $C_1 = -w_0/24$.

$$y(x) = -[w_0/(24EI)] x(1 - 2x^2 + x^3)$$

The negative sign means the deflection is downward if upward is positive. If the y -axis is taken downward as in the sketch, the downward deflection magnitude is positive:

$$\delta(x) = [w_0/(24EI)] x(1 - 2x^2 + x^3), \quad 0 \leq x \leq 1$$

Final answer

Deflection magnitude downward: $\delta(x) = \frac{w_0}{24EI} x(1-2x^2+x^3)$, $0 \leq x \leq 1$. With upward-positive y , $y = -\delta$.

Common mistake / exam note

The sign of beam deflection depends on the chosen y -axis and moment convention. State the convention; the physical downward magnitude is the same.

7(b) Prove the direction-cosine matrix is orthogonal

The rows are the direction cosines of three mutually perpendicular lines:

$$A = \begin{bmatrix} l_1, m_1, n_1 \\ l_2, m_2, n_2 \\ l_3, m_3, n_3 \end{bmatrix}$$

Because each row represents a unit direction:

$$l_r^2 + m_r^2 + n_r^2 = 1, \quad r = 1, 2, 3$$

Because the three lines are mutually perpendicular, the dot product of any two different direction rows is zero:

$$l_1 l_2 + m_1 m_2 + n_1 n_2 = 0, \text{ and similarly for pairs } (1,3) \text{ and } (2,3)$$

Therefore, when we multiply A by A^T , diagonal entries are 1 and off-diagonal entries are 0:

$$A A^T = I$$

$$\text{Hence } A^{-1} = A^T$$

Final answer

A is orthogonal because its rows are an orthonormal set, so $A A^T = I$.

7(c) Eigenvalues of $A = \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix}$

$$\det(A - \lambda I) = (0.5 - \lambda)^2 - 0.25$$

$$= \lambda^2 - \lambda = \lambda(\lambda - 1) = 0$$

$$\text{Eigenvalues: } \lambda = 0 \text{ and } \lambda = 1$$

The largest eigenvalue is 1. Solve $(A - I)v = 0$:

$$\begin{bmatrix} -0.5 & 0.5 \\ 0.5 & -0.5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}^T = 0 \rightarrow x = y$$

Final answer

Eigenvalues are 1 and 0. An eigenvector for the largest eigenvalue 1 is $(1, 1)^T$; a unit eigenvector is $(1/\sqrt{2})(1, 1)^T$.

SET 8 - Homogeneous Systems, Transition Matrix and Matrix Inverse

What this set tests

8(a) Solve $AX=0$; then solve $(BA)X=0$ for the given transition matrix B and explain whether the solution space changes.

8(b) Find A^{-1} and use $Ax=H$ to determine x .

Basics you need: A homogeneous system $AX=0$ has non-trivial solutions when A is singular. If B is invertible, $BAx=0$ and $Ax=0$ have exactly the same solution space. For $Ax=H$ with $\det(A)$ nonzero, $x=A^{-1}H$.

Reusable procedure

For $AX=0$: row-reduce $A \rightarrow$ choose the free variable $\rightarrow$ write the parametric vector solution. For $(BA)X=0$: check that B is invertible, then multiply by B^{-1} . For $Ax=H$: compute A^{-1} and multiply by H .

8(a) Solve $AX=0$ and $(BA)X=0$

$$A = \begin{bmatrix} 2 & -1 & 3 \\ 3 & 2 & 1 \\ 1 & 3 & -2 \end{bmatrix}$$

Use $R_2 \rightarrow 2R_2 - 3R_1$ and $R_3 \rightarrow 2R_3 - R_1$. Both become $[0, 7, -7]$, so the independent equations are $2x - y + 3z = 0$ and $y - z = 0$.

Let $z = t$. Then $y = t$ and $2x + 2t = 0$, so $x = -t$.

$$X = t \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}^T, \text{ where } t \text{ is any real number.}$$

The transition matrix $B = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$ is a permutation matrix, hence $B^{-1} = B$ and $\det(B)$ is nonzero. Therefore $(BA)X=0$ if and only if $AX=0$.

Final answer: $X = t \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}^T$ for both systems. The solution space is unchanged because B is invertible.

8(b) Find A^{-1} and solve $Ax=H$

$$A = \begin{bmatrix} 1 & 2 & -1 \\ -1 & 1 & 2 \\ 2 & -1 & 1 \end{bmatrix}, \quad H = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}^T$$

$\det(A) = 14$, so A is nonsingular and its inverse exists.

$$A^{-1} = (1/14) \begin{bmatrix} 3 & -1 & 5 \\ 5 & 3 & -1 \\ -1 & 5 & 3 \end{bmatrix}$$

$$x = A^{-1}H = (1/14) \begin{bmatrix} 8 \\ 4 \\ 2 \end{bmatrix}^T = \begin{bmatrix} 4/7 \\ 2/7 \\ 1/7 \end{bmatrix}^T$$

Final answer: $A^{-1} = (1/14) \begin{bmatrix} 3 & -1 & 5 \\ 5 & 3 & -1 \\ -1 & 5 & 3 \end{bmatrix}$ and $x = \begin{bmatrix} 4/7 \\ 2/7 \\ 1/7 \end{bmatrix}^T$.

Rapid Revision - Procedures to Memorize

Topic	Recipe
T-N and curvature	$T = r'/ r' $; $N = T'/ T' $; $\kappa = T' / r' $; $\rho = 1/\kappa$; normal plane uses T as normal.
Solenoidal / rotational	$\text{div } F = 0 \rightarrow$ solenoidal. $\text{curl } F \neq 0 \rightarrow$ rotational.
Maximum directional derivative	Compute $\text{grad } f$. Maximum magnitude $= \text{grad } f $ and direction $= \text{grad } f$.
Solenoidal product	$\text{div}(\phi A) = \text{grad } \phi \cdot A + \phi \text{div } A$.
Work along curve	$r(t) \rightarrow r'(t) \rightarrow F(r(t)) \cdot r'(t) \rightarrow$ integrate.
Surface flux $z=f(x,y)$	Use $(-f_x, -f_y, 1) dx dy$ for upward orientation.
Piecewise line integral	Parameterize each segment separately and add the integrals.
Divergence volume integral	Compute div first, then integrate over the box.
Rank / RREF	Eliminate $\rightarrow$ count pivots $\rightarrow$ continue until pivot columns are unit columns.
System classification	$\det \neq 0$ unique. If $\det = 0$ compare rank A and rank augmented.
Laplace t^n property	$L\{t^n f\} = (-1)^n F^{(n)}(s)$.
Periodic Laplace	One-period integral divided by $1 - e^{-sT}$.

Convolution	Product in s-domain $\rightarrow \int_0^t f(\tau)g(t-\tau)d\tau$.
Laplace ODE	Transform $\rightarrow$ insert initial conditions $\rightarrow$ solve $Y(s)$ $\rightarrow$ inverse transform.
Integral equation	Recognize convolution $\rightarrow$ transform $\rightarrow$ isolate unknown transform.
Beam deflection	Find reactions $\rightarrow M(x)$ $\rightarrow$ integrate $EI y''=M$ twice $\rightarrow$ apply support conditions.
Orthogonal matrix	Rows/columns orthonormal $\rightarrow AA^T=I$.
Eigenvalues	$\det(A-\lambda I)=0$ $\rightarrow$ solve eigenvectors from $(A-\lambda I)v=0$.
Homogeneous systems / transition matrix	Row-reduce $AX=0$. If B is invertible, $\text{null}(BA)=\text{null}(A)$. For $Ax=H$ with $\det(A)\neq 0$, $x=A^{-1}H$.

Final answers at a glance - 2022

Question	Answer
1(a)	$T=(-\sin t, \cos t, 0)$, $N=(-\cos t, -\sin t, 0)$, $\rho=2$; at $t=\pi/2$ normal plane $x=0$.
1(b)	Both solenoidal and rotational if $\text{div } F=0$ and $\text{curl } F \neq 0$.
2(a)	Expected $+y$: $a=b=16, c=4$; x -direction derivative at $(1, -1, 2)=24$.
2(b)	$\text{div } A=x(1+x^2) \neq 0$; $\text{div}(xyz^2 A)=0$.
2(c)	Work=303.
3(a)	Flux=8 for positive- z /outward normal; opposite orientation gives -8.
3(b)	Line integral=11.
3(c)	176.
4(a)	rank=3; RREF= $\begin{bmatrix} 1 & 0 & 0 & -5/6 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 5/3 \end{bmatrix}$; normal= $[3 0]$.
4(b)	$k \neq 2$ unique $(0, 1/3, 0)$; $k=2$ infinite: $(t, 1/3-4t/3, t)$; no no-solution value.
5(a)	$L\{t^3 \sin t\}=24s(s^2-1)/(s^2+1)^4$; integral with $e^{-t}=0$.
5(b)	$[3(1-e^{-2s})/s^2 - 6e^{-4s}/s]/(1-e^{-4s})$.
5(c)	$t-2+(t+2)e^{-t}$.
6(a)	$e^t-1-(1/2)\sin 2t$.
6(b)	$y=e^{-2t}(2t^2+2t-1)$.
6(c)	$F(t)=1+2t$.
7(a)	Downward magnitude $w_0 x(1-2x^2+x^3)/(24EI)$.
7(b)	$AA^T=I$, so A is orthogonal.

7(c)	Eigenvalues 1,0; largest-eigenvalue eigenvector proportional to (1,1).
8(a)	$X=t[-1,1,1]^T$ for both $AX=0$ and $(BA)X=0$; solution space unchanged because B is invertible.
8(b)	$A^{-1} = (1/14)[[3,-1,5],[5,3,-1],[-1,5,3]]$, $x = [4/7, 2/7, 1/7]^T$.

Exam Strategy for the 2022 Paper

If your basics are weak, prioritize in this order

1. Sets 4-6: row operations and Laplace questions are highly procedural and become easier with a fixed recipe.
2. Set 2: learn gradient, divergence, and work integral formulas; these repeat in many years.
3. Set 3: practice choosing limits for a triangle, a path made of segments, and a rectangular box.
4. Set 7: memorize the standard simply-supported UDL beam result only after understanding the two integrations.
5. Set 1: T, N, curvature, and the normal plane are easiest when you always follow the same derivative sequence.

Final checking habit

Write the governing formula before substitution.

For vector integrals, state the path/surface orientation and limits.

For matrices, show row operations clearly.

For Laplace ODEs, substitute initial values immediately after transforming.

If the printed question appears to contain a typo, state the mathematically meaningful interpretation instead of silently guessing.